Riemann Sums and Definite Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Approximating area with Riemann sums

Estimate $\int_0^4 x^2\,dx$ using a right-endpoint Riemann sum with $n = 4$ subintervals.

$\Delta x = 1$; right endpoints $1, 2, 3, 4$: sum $= 1(1) + 1(4) + 1(9) + 1(16) = 30$.

2 [Foundational] • Target: Approximating area with Riemann sums

Estimate $\int_1^5 (2x + 1)\,dx$ using a left-endpoint Riemann sum with $n = 4$ subintervals.

$\Delta x = 1$; left endpoints $1, 2, 3, 4$: values $3, 5, 7, 9$: sum $= 24$.

3 [Foundational] • Target: The definite integral and its properties

Given $\int_0^3 f(x)\,dx = 8$ and $\int_3^7 f(x)\,dx = -2$, find:

a $\int_0^7 f(x)\,dx = 6$ **b** $\int_7^3 f(x)\,dx = 2$ **c** $\int_0^3 5f(x)\,dx = 40$

(a) $= 6$; (b) $= 2$; (c) $= 40$

4 [Foundational] • Target: The definite integral and its properties

The graph of $f(x) = 4 - x^2$ is shown, together with the region between the curve and the x-axis on $[0, 2]$. Evaluate $\int_0^2 (4 - x^2)\,dx$ exactly.

$\left[4x - \frac{x^3}{3}\right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3}$.

5 [Foundational] • Target: The definite integral and its properties

Evaluate $\int_{-2}^2 (x^3 - 4x)\,dx$ by recognizing the integrand's symmetry, without computing an antiderivative.

The integrand is odd ($f(-x) = -f(x)$), and the interval is symmetric about 0, so the integral is 0.

6 [Foundational] • Target: Midpoint and trapezoidal approximations

Estimate $\int_0^4 x^2\,dx$ using a midpoint Riemann sum with $n = 4$ subintervals, and compare to the exact value $\frac{64}{3} \approx 21.33$.

$\Delta x = 1$; midpoints $0.5, 1.5, 2.5, 3.5$: sum $= 0.25 + 2.25 + 6.25 + 12.25 = 21$. Close to the exact value 21.33.

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7 [Foundational] • Target: Midpoint and trapezoidal approximations

Estimate $\int_1^5 \frac{1}{x}\,dx$ using the trapezoidal rule with $n = 4$ subintervals, to three decimal places.

$\Delta x = 1$, values at $1, 2, 3, 4, 5$: $1, 0.5, 0.333, 0.25, 0.2$. Trapezoidal:
$\frac{1}{2}(1 + 2(0.5 + 0.333 + 0.25) + 0.2) = \frac{1}{2}(1.2 + 2.166) \approx 1.683$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Signed area

The graph of $f(x) = x - 2$ crosses the x-axis at $x = 2$. Evaluate $\int_0^4 (x - 2)\,dx$ and explain why the result does not equal the total area between the curve and the axis.

$\left[\frac{x^2}{2} - 2x\right]_0^4 = (8 - 8) - 0 = 0$. The integral gives signed (net) area: the region below the axis on $[0, 2]$ cancels the region above it on $[2, 4]$, even though both have positive area.

9 [Intermediate] • Target: Signed area

Find the total (unsigned) area between $f(x) = x - 2$ and the x-axis on $[0, 4]$, splitting the integral at the sign change $x = 2$.

$\left|\int_0^2 (x - 2)\,dx\right| + \left|\int_2^4 (x - 2)\,dx\right|$. $\int_0^2 (x - 2)\,dx = \left[\frac{x^2}{2} - 2x\right]_0^2 = (2 - 4) - 0 = -2$, so $|-2| = 2$.
$\int_2^4 (x - 2)\,dx = \left[\frac{x^2}{2} - 2x\right]_2^4 = (8 - 8) - (2 - 4) = 0 - (-2) = 2$. Total area $= 2 + 2 = 4$.

10 [Intermediate] • Target: Comparing Riemann sum estimates

For the increasing function $f(x) = x^2$ on $[0, 4]$, will the left-endpoint sum overestimate or underestimate the true area? Justify your answer.

Underestimate — since f is increasing, the left endpoint of each subinterval gives the smallest function value on that subinterval, so each rectangle sits below the curve.

11 [Intermediate] • Target: Comparing Riemann sum estimates

Using the right-endpoint result (30) and left-endpoint result (24, computed earlier for a different function) is not directly comparable — instead, compare the right-endpoint estimate (30) for $\int_0^4 x^2 dx$ to the exact value $\frac{64}{3} \approx 21.33$, and explain the direction of the error.

The right-endpoint estimate (30) overestimates the true value (≈ 21.33), consistent with $f(x) = x^2$ being increasing — right endpoints give the largest function value on each subinterval, so the rectangles sit above the curve.

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12 [Intermediate] • Target: Riemann sums from a table of values

A table gives $f(0) = 2$, $f(1) = 5$, $f(2) = 9$, $f(3) = 14$, $f(4) = 20$. Use a right-endpoint Riemann sum with these 4 equal subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

$R_4 = 1[f(1) + f(2) + f(3) + f(4)] = 5 + 9 + 14 + 20 = 48.$

13 [Intermediate] • Target: The definite integral as a limit of Riemann sums

Write the formal limit definition of the definite integral $\int_a^b f(x)\,dx$ in terms of a Riemann sum, and explain the role of $n \to \infty$.

$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$, where $\Delta x = \frac{b-a}{n}$. As $n \to \infty$, the rectangles become infinitely thin, and the sum converges to the exact signed area under the curve, regardless of which sample point x_i^* (left, right, midpoint) is used within each subinterval.

14 [Intermediate] • Target: Riemann sums with sigma notation

Write the right-endpoint Riemann sum R_n for $\int_1^3 x^2\,dx$ using sigma notation, with $\Delta x = \frac{2}{n}$ and $x_i = 1 + i\Delta x$. Do not evaluate the sum.

$R_n = \sum_{i=1}^{n} f(x_i)\Delta x = \sum_{i=1}^{n} \left(1 + \frac{2i}{n}\right)^2 \cdot \frac{2}{n}.$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Riemann sums with sigma notation • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Estimate $\int_0^6 f(x)\,dx$ using a midpoint Riemann sum with $n = 3$ subintervals, given $f(1) = 4$, $f(3) = 10$, $f(5) = 16$.

$\Delta x = 2$; midpoints are exactly $1, 3, 5$: sum $= 2(4 + 10 + 16) = 60$.

16 [Challenging] • Target: Comparing approximation methods on a concave function • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

For a concave-down function on $[a, b]$, which is larger: the trapezoidal estimate or the midpoint estimate? Justify briefly, and state which one overestimates the true area.

For concave-down f, the trapezoids lie below the curve (underestimate), while midpoint rectangles' tops cross above the curve on average (overestimate). So the midpoint estimate is larger, and the trapezoidal estimate underestimates the true area.

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17 [Challenging] • Target: Signed area • Cross-Lesson: 15.01 Double Integrals

For $f \geq 0$ continuous on $[a, b]$, FTC connects $\int_a^b f(x)dx$ to the area under the curve because Riemann sums $\sum f(x_i^*)\Delta x$ converge to the integral as:.

Final Answer: $n \to \infty$ (partition width $\to 0$). The definite integral is defined as the limit of Riemann sums as the number of subintervals $n \to \infty$.

18 [Challenging] • Target: Comparing Riemann sum estimates • Cross-Lesson: 16.01 Vector Fields and Line Integrals

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$? Show your complete algebraic work.

Final Answer: 17. The Fundamental Theorem of Calculus relates the definite integral of a function to its antiderivative. Geometrically, the definite integral $\int_a^b f(x)\,dx$ represents the net signed area between the curve $f(x)$ and the x-axis from $x = a$ to $x = b$. Areas above the x-axis are positive, and areas below the x-axis are negative. The problem states that the area of the shaded region, representing $\int_a^c f(x)\,dx$, is 12 square units. This implies that this region contributes positivel.

19 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 2.01 Evaluating Limits

Consider the function $f(x) = x^2$. What is the value of the definite integral $\int_0^3 f(x)dx$? Show your complete algebraic work.

Final Answer: 9. The integral of $f(x) = x^2$ is $\frac{x^3}{3}$. Evaluating this from 0 to 3 gives: $\left[\frac{x^3}{3}\right]_0^3 = \frac{3^3}{3} - \frac{0^3}{3} = \frac{27}{3} - 0 = 9$. Therefore, the value of the definite integral is 9.

20 [Challenging] • Target: The definite integral as a limit of Riemann sums • Cross-Lesson: 2.02 Continuity

Evaluate the improper integral $\int_1^\infty \frac{\ln(x)}{x^4}\,dx$.

Final Answer: $\frac{1}{9}$. Write it as a limit, e.g. $\lim_{b\to\infty} \int_1^b \frac{\ln(x)}{x^4}\,dx$, integrate, and take the limit. The integral converges to $\frac{1}{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Left and Right Riemann Sums

Approximate $\int_0^4 x^2\,dx$ using a Left Riemann sum with $n = 4$ equal subintervals.

a. 14 ✓

b. 30

c. 21

d. 16

$\Delta x = \frac{4-0}{4} = 1$. Grid points: $0, 1, 2, 3, 4$. Left sum
$= 1 \cdot [f(0) + f(1) + f(2) + f(3)] = 1 \cdot [0 + 1 + 4 + 9] = 14$.

22 [Foundational] • Target: Left and Right Riemann Sums

Approximate $\int_0^4 x^2\,dx$ using a Right Riemann sum with $n = 4$ equal subintervals.

a. 30 ✓

b. 14

c. 22

d. 28

Right sum $= 1 \cdot [f(1) + f(2) + f(3) + f(4)] = 1 \cdot [1 + 4 + 9 + 16] = 30$.

23 [Foundational] • Target: Left and Right Riemann Sums

For a strictly increasing function $f(x)$ on $[a, b]$, which inequality correctly relates the exact integral I to the Left (L_n) and Right (R_n) Riemann sums?

a. $L_n < I < R_n$ ✓

b. $R_n < I < L_n$

c. $I < L_n < R_n$

d. $L_n = I = R_n$

For an increasing function, rectangles evaluated at the left endpoints lie below the curve ($L_n < I$) and right endpoints lie above the curve ($I < R_n$).

24 [Foundational] • Target: Midpoint and Trapezoidal Approximations

Approximate $\int_1^5 \frac{1}{x}\,dx$ using the Midpoint Rule with $n = 2$ equal subintervals.

a. $\frac{1}{2} + \frac{1}{4} = 0.75 \approx 1.5$ **is wrong; width=2:** $2(1/2 + 1/4) = 1.5$ ✓

b. 1.5

c. 1.2

d. 1.33

$\Delta x = \frac{5-1}{2} = 2$. Subintervals: $[1, 3]$ and $[3, 5]$. Midpoints are $m_1 = 2$ and $m_2 = 4$. Midpoint sum
$= 2 \cdot [f(2) + f(4)] = 2(1/2 + 1/4) = 2(3/4) = 1.5$.

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25 [Foundational] • Target: Midpoint and Trapezoidal Approximations

Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^2 x^3\,dx$.

a. 4.25 ✓

b. 4.00

c. 3.75

d. 4.50

$\Delta x = 0.5$. Grid: $0, 0.5, 1, 1.5, 2$. Values: $0, 0.125, 1, 3.375, 8$.

$T_4 = \frac{0.5}{2}[0 + 2(0.125 + 1 + 3.375) + 8] = 0.25[0 + 9 + 8] = 0.25(17) = 4.25$.

26 [Foundational] • Target: Midpoint and Trapezoidal Approximations

If $f''(x) > 0$ (concave up) on $[a, b]$, how does the Trapezoidal approximation T_n compare to the exact integral I?

a. $T_n > I$ **(overestimate)** ✓

b. $T_n < I$ (underestimate)

c. $T_n = I$

d. Cannot be determined without $f'(x)$

When f is concave up, secant lines connecting adjacent points on the curve lie entirely above the curve, so trapezoids overestimate the area: $T_n > I$.

27 [Foundational] • Target: Definite Integral as the Limit of a Riemann Sum

Express the limit $\lim_{n\to\infty} \sum_{i=1}^{n} \left(2 + \frac{3i}{n}\right)^4 \frac{3}{n}$ as a definite integral.

a. $\int_2^5 x^4\,dx$ ✓

b. $\int_0^3 (2 + x)^4\,dx$

c. $\int_2^5 (2 + 3x)^4\,dx$

d. $\int_0^1 (2 + 3x)^4\,dx$

Here $\Delta x = 3/n$ and $x_i = 2 + i\Delta x$. Lower limit $a = 2$, upper limit $b = 2 + 3 = 5$. Integrand is x^4. Integral is $\int_2^5 x^4\,dx$.

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28 [Foundational] • Target: Definite Integral as the Limit of a Riemann Sum

Evaluate $\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} \frac{i^2}{n^2}$.

a. 1/3 ✓ **b.** 1/2

c. 1/4 **d.** 1

This corresponds to $\int_0^1 x^2\,dx = [x^3/3]_0^1 = 1/3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Definite Integral as the Limit of a Riemann Sum

Convert $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{n+i}$ into a definite integral on $[0, 1]$ and evaluate it.

a. $\ln 2$ ✓ **b.** 1

c. $\ln 3$ **d.** 1/2

Factor out $1/n$: $\lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} \frac{1}{1+i/n} = \int_0^1 \frac{1}{1+x}\,dx = [\ln(1+x)]_0^1 = \ln 2$.

30 [Intermediate] • Target: Geometric Evaluation of Definite Integrals

Evaluate $\int_{-3}^{3} \sqrt{9-x^2}\,dx$ using geometry.

a. $\frac{9\pi}{2}$ ✓ **b.** 9π

c. $\frac{9\pi}{4}$ **d.** 18

The graph of $y = \sqrt{9-x^2}$ is the upper semicircle of radius $r = 3$ centered at the origin. Area $= \frac{1}{2}\pi r^2 = \frac{1}{2}\pi(9) = \frac{9\pi}{2}$.

31 [Intermediate] • Target: Geometric Evaluation of Definite Integrals

Evaluate $\int_0^4 (2x-3)\,dx$ by computing net signed area.

a. 4 ✓ **b.** 8

c. 0 **d.** -2

Triangles: below axis on $[0, 1.5]$: base 1.5, height 3, area $-(1.5 \times 3)/2 = -2.25$. Above axis on $[1.5, 4]$: base 2.5, height 5, area $(2.5 \times 5)/2 = 6.25$. Net $= 6.25 - 2.25 = 4$.

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32 [Intermediate] • Target: Geometric Evaluation of Definite Integrals

If $\int_1^5 f(x)\,dx = 7$ and $\int_3^5 f(x)\,dx = 4$, find $\int_1^3 f(x)\,dx$.

a. 3 ✓ **b.** 11

c. -3 **d.** $7/4$

By additivity of intervals: $\int_1^5 f = \int_1^3 f + \int_3^5 f \Rightarrow 7 = \int_1^3 f + 4 \Rightarrow \int_1^3 f = 3$.

33 [Intermediate] • Target: Geometric Evaluation of Definite Integrals

Evaluate the improper integral $\int_0^\infty \frac{1}{(x+4)^4}\,dx$.

a. $\frac{1}{192}$ ✓ **b.** $\frac{1}{96}$

c. ∞ **d.** $\frac{193}{192}$

Write it as a limit, e.g. $\lim_{b\to\infty} \int_0^b \frac{1}{(x+4)^4}\,dx$, integrate, and take the limit. The integral converges to $\frac{1}{192}$.

34 [Intermediate] • Target: Approximating area with Riemann sums

Evaluate the improper integral $\int_1^\infty \frac{\ln(x)}{x^3}\,dx$.

a. $\frac{1}{2}$ **b.** ∞

c. $\frac{1}{4}$ ✓ **d.** $\frac{5}{4}$

Write it as a limit, e.g. $\lim_{b\to\infty} \int_1^b \frac{\ln(x)}{x^3}\,dx$, integrate, and take the limit. The integral converges to $\frac{1}{4}$.

35 [Intermediate] • Target: The definite integral and its properties

Evaluate the improper integral $\int_0^1 x^3 \ln(x)\,dx$.

a. $-\frac{1}{8}$ **b.** $-\frac{1}{16}$ ✓

c. ∞ **d.** $\frac{15}{16}$

Write it as a limit, e.g. $\lim_{a\to 0^+} \int_a^1 x^3 \ln(x)\,dx$, integrate, and take the limit. The integral converges to $-\frac{1}{16}$.

36 [Intermediate] • Target: Midpoint and trapezoidal approximations

Evaluate the improper integral $\int_0^1 x^2 \ln(x)\,dx$.

a. $-\frac{2}{9}$ **b.** $-\frac{1}{9}$ ✓

c. ∞ **d.** $\frac{8}{9}$

Write it as a limit, e.g. $\lim_{a\to 0^+} \int_a^1 x^2 \ln(x)\,dx$, integrate, and take the limit. The integral converges to $-\frac{1}{9}$.

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37 [Intermediate] • Target: Signed area

For $|x| < 5$, which function equals $\sum_{n=0}^{\infty} \left(\frac{x}{5}\right)^n$?

a. $\frac{1}{1-x}$

b. $\frac{5}{5-x}$ ✓

c. $\frac{5}{x+5}$

d. $\ln(x+1)$

Geometric series with ratio x/5: the sum is $\frac{5}{5-x}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Comparing Riemann sum estimates • Cross-Lesson: 16.01 Vector Fields and Line Integrals

For $|x| < 3$, which function equals $\sum_{n=0}^{\infty} \left(\frac{x}{3}\right)^n$?

a. $\frac{1}{1-x}$

b. $\frac{3}{x+3}$

c. $\frac{3}{3-x}$ ✓

d. $\ln(x+1)$

Geometric series with ratio x/3: the sum is $\frac{3}{3-x}$.

39 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 2.01 Evaluating Limits

For $|x| < 2$, which function equals $\sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n$?

a. $\frac{1}{1-x}$

b. $\frac{2}{x+2}$

c. $\ln(x+1)$

d. $\frac{2}{2-x}$ ✓

Geometric series with ratio x/2: the sum is $\frac{2}{2-x}$.

40 [Challenging] • Target: The definite integral as a limit of Riemann sums • Cross-Lesson: 2.02 Continuity

For $|x| < 4$, which function equals $\sum_{n=0}^{\infty} \left(\frac{x}{4}\right)^n$?

a. $\frac{1}{1-x}$

b. $\frac{4}{x+4}$

c. $\frac{4}{4-x}$ ✓

d. $\ln(x+1)$

Geometric series with ratio x/4: the sum is $\frac{4}{4-x}$.

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41 [Challenging] • Target: Riemann sums with sigma notation • Cross-Lesson: 3.01 The Derivative from First Principles

Bessel function $J_0(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{(m!)^2}\left(\frac{x}{2}\right)^{2m}$. Radius of convergence:

a. 1

b. ∞ ✓

c. π

d. 2

Ratio test: $L = 0 \Rightarrow R = \infty$.

42 [Challenging] • Target: Comparing approximation methods on a concave function • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

For iterated integral $\int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy \right) dx$: the inner integral is done first, then the outer. True or false?

a. False — outer first

b. **True** ✓

c. Only for separable functions

d. Depends on Fubini

Inner integral is evaluated first.

43 [Challenging] • Target: Left and Right Riemann Sums • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{2^{-n}(x-1)^n}{n^2}$.

a. $(-1, 3)$

b. $[-1, 3]$ ✓

c. $[-1, 3)$

d. $(-1, 3]$

The radius is 2 (ratio test). At $x = 3$ the series is $\sum_{n=1}^{\infty} \frac{1}{n^2}$; at $x = -1$ it is $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$. Checking each endpoint gives $[-1, 3]$.

44 [Challenging] • Target: Midpoint and Trapezoidal Approximations • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{4^{-n}(x-3)^n}{n^2}$.

a. $(-1, 7)$

b. $[-1, 7]$ ✓

c. $[-1, 7)$

d. $(-1, 7]$

The radius is 4 (ratio test). At $x = 7$ the series is $\sum_{n=1}^{\infty} \frac{1}{n^2}$; at $x = -1$ it is $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$. Checking each endpoint gives $[-1, 7]$.

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45 [Challenging] • Target: Definite Integral as the Limit of a Riemann Sum • Cross-Lesson: 6.01 Related Rates

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x+3)^n}{n}$.

a. $(-4, -2)$

b. $[-4, -2]$

c. $(-4, -2]$

d. $[-4, -2)$ ✓

The radius is 1 (ratio test). At $x = -2$ the series is $\sum_{n=1}^{\infty} \frac{1}{n}$; at $x = -4$ it is $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$. Checking each endpoint gives $[-4, -2)$.

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